

Human Classifier Simulation

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can construct a classifier with a stated input format, selected rule and recorded output.

I can trace valid, boundary and invalid inputs through different paths.

I can distinguish an input check from evidence that a prediction is correct.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 10: Human Classifier Simulation

A classifier applies a learned or designed boundary to features.



THINKER MISSION

Train classmates on labeled creatures and compare their rules.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

A fitted rule predicts tall at a height of at least 6 cm and short otherwise. Predict at 5 and 6 cm. What should you check before applying it to a blank height?

This week's end goal

By the end, I can simulate a classifier with explicit requirements and missing-value behavior.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use human classifier simulation to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Your task is to build a paper classifier that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers. Use the threshold of 6 centimetres selected after adding the height 5 short example in Week 13. Name this selected model Tower1.

Write an input contract: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts. It is not a claim that every accepted height appeared in fitting or that towers outside this range cannot exist.

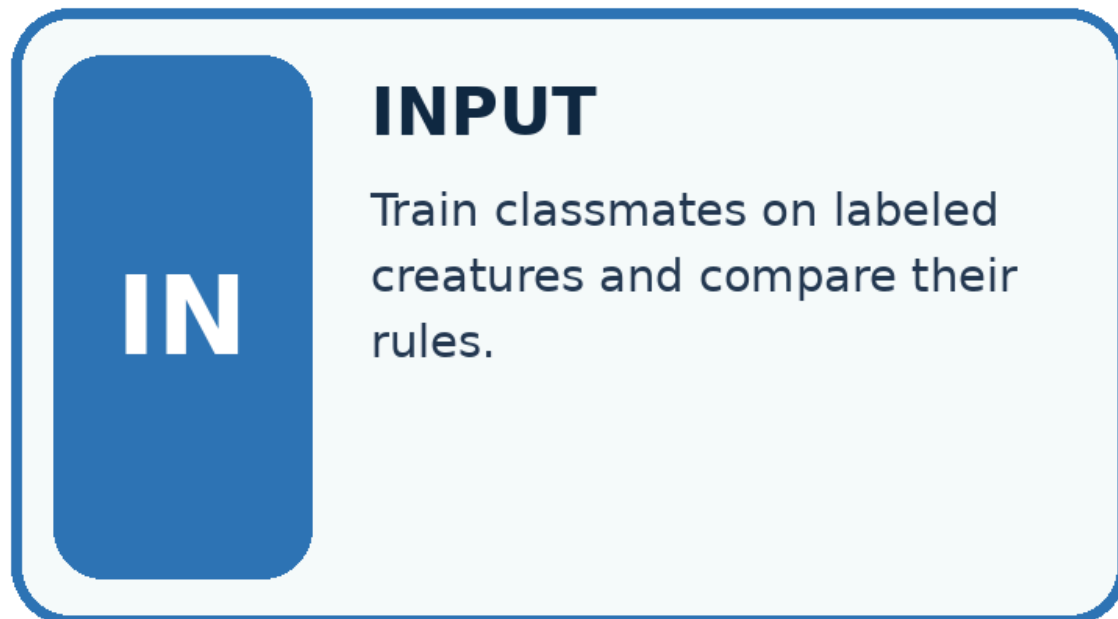
A complete classifier needs instructions for checking the input, applying Tower1 when the input is accepted, and recording the outcome. It must also say what happens when the input fails a check. The threshold alone does not provide these surrounding instructions.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



PROCESS

A classifier applies a learned or designed boundary to features.



OUT

OUTPUT

Train classmates on labeled creatures and compare their rules.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Three records through the Tower1 classifier

Input record · Check and rule · Recorded outcome

Input record: 5 cm

Check and rule: Accepted; 5 is below 6

Recorded outcome: Tower1 predicts short

Input record: 6 cm

Check and rule: Accepted; 6 is at least 6

Recorded outcome: Tower1 predicts tall

Input record: Height blank

Check and rule: Missing height; stop before rule

Recorded outcome: Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10. If any check fails, record no prediction and the reason, then stop processing that record. Do not continue into the threshold comparison.

If all checks pass, compare the height with 6. At least 6 gives the prediction tall; otherwise give short. Record the original input, model version Tower1 and the prediction. For a rejected record, record the original input, Tower1, no prediction and a reason. Tower1 identifies the intended model even when its threshold was not used.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected everyday case



BOUNDARY
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Supply M's pages as 101.

Prediction: M is now rejected rather than awaiting page review.

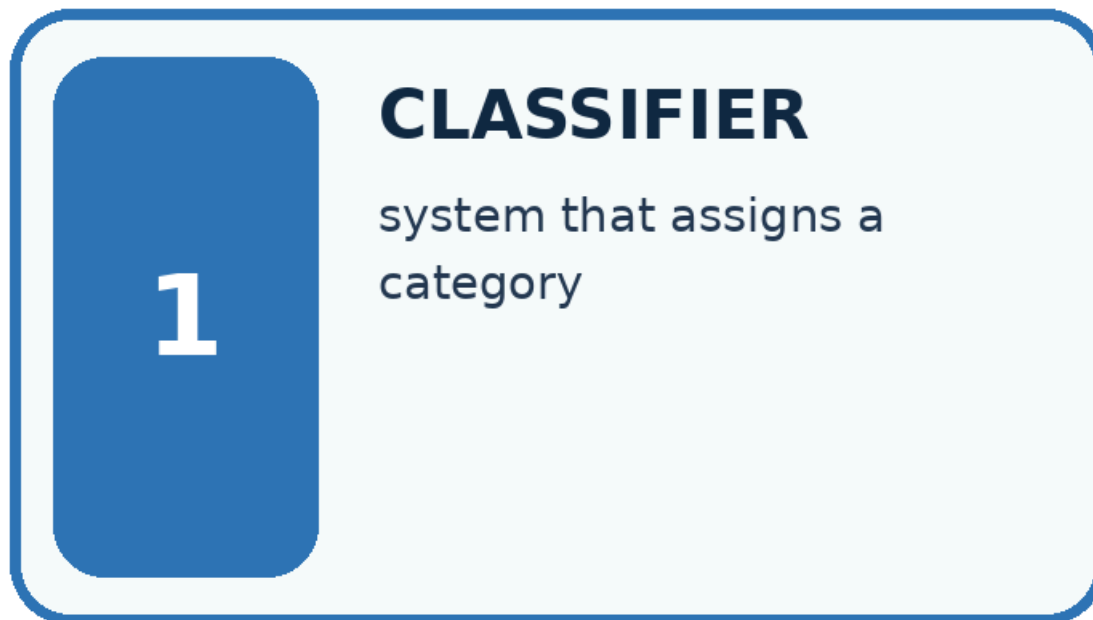
Reason: The required field is known but fails the maximum-length condition.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

BOUNDARY

rule separating categories

3

TRAINING EXAMPLE

labeled case used to create
behavior

4

PREDICTION

category assigned to a new case

Word · My own example · Not this / boundary

Word: classifier

My own example: A rule or model that assigns an input to a category.

Not this / boundary: _____

Word: boundary

My own example: A dividing condition between categories, actions or allowed behaviours.

Not this / boundary: _____

Word: training example

My own example: An input and any associated label used to fit a model.

Not this / boundary: _____

Word: prediction

My own example: An estimated category, value or outcome that must be distinguished from established fact.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Input contract

Meaning: The format and conditions an input must satisfy for a procedure to accept it

Word: Validation

Meaning: Checking whether an input satisfies the stated contract

Word: No prediction

Meaning: An explicit outcome when the classifier does not assign a category

Word: Model version

Meaning: A name that identifies the particular model used for a prediction

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 10 • WORKED EVIDENCE

- 1 Examples suggest: striped AND small → glim.
- 2 Learners infer a boundary from labeled cards.
- 3 New case: striped, small creature → predict glim.
- 4 Different inferred rules may match training yet disagree on a new case.

1. Examples suggest: striped AND small → glim.
2. Learners infer a boundary from labeled cards.
3. New case: striped, small creature → predict glim.
4. Different inferred rules may match training yet disagree on a new case.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Input · Contract result · Outcome

Input: 1 cm

Contract result: Accepted at lower endpoint

Outcome: Predict short

Input: 9 cm

Contract result: Accepted

Outcome: Predict tall

Input: 10 cm

Contract result: Accepted at upper endpoint

Outcome: Predict tall

Input: 0 cm

Contract result: Outside 1 through 10

Outcome: No prediction

Input: 60 mm

Contract result: Unit is not cm

Outcome: No prediction

Input: 4.5 cm

Contract result: Not a whole number

Outcome: No prediction

A valid 9 cm input receives a tall prediction, but the input check does not establish that the prediction matches its comparison label. Validation checks the contract. Evaluation checks predictions against appropriate evidence. Keep these questions separate.

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step. Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Human Classifier Simulation

A fictional librarian wants illustrated mystery books with at most 100 pages.

The facts we will use

Written rule: choose only genre mystery AND illustrated True AND pages $\leq$ 100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

Human Classifier Simulation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read all conditions

Three requirements must pass, not just a high preference score.



2. Trace J

Mystery, illustrated and exactly 100 pages: choose.



3. Trace K and L

K fails illustration; L fails genre: do not choose.



4. Trace M

Missing pages means review rather than assuming it is short.

CHANGE ONE THING

Supply M's pages as 101.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read all conditions

What happens and why: Three requirements must pass, not just a high preference score.

Step: 2. Trace J

What happens and why: Mystery, illustrated and exactly 100 pages: choose.

Step: 3. Trace K and L

What happens and why: K fails illustration; L fails genre: do not choose.

Step: 4. Trace M

What happens and why: Missing pages means review rather than assuming it is short.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

What this example does not establish

This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: A classifier remembers the correct label for every possible future item. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

The classifier reports a prediction or no prediction. It does not decide what a person must do with a tower. A separate classroom procedure might ask someone to remeasure a rejected input or compare a prediction with a recorded label. Write that action separately from the predicted category.

No prediction is not a third tower size. It records that this procedure did not assign short or tall. Count and investigate such records when evaluating the whole system; hiding them could make a system look more useful than it is.

A classifier accepts 9 cm and predicts tall. A learner says the input passed, so the label is definitely right. Correct the claim. Then explain why reporting short for a blank height would violate the Tower1 procedure

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Write the procedure as numbered steps or connected boxes with labelled branches. Give it to a partner who follows only what you wrote. The partner should record each check and stop point, including equality at the threshold and at both accepted range endpoints.

Construct Tower1 in your own words. Have a partner trace 1 cm, 6 cm, 10 cm, 11 cm and a blank height. Record each outcome and any instruction you had to clarify. Explain why the accepted 10 cm case still needs separate evidence of correctness

A partner finding an unclear branch gives useful design feedback. Revise the instruction and retrace the affected examples. Keep the model name and threshold visible so a change to the fitted rule is not confused with a correction to an instruction.

Practise with less help: Construct both outcome paths

Use workbook W2 for the connected example “Constructing a classifier”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Constructing a classifier. Proposed explanation: “Any numeric input must receive a category”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why are J and M initially treated differently even though both are illustrated mysteries?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Reveal six labeled creature cards, hide labels on four new cards, record each learner's rule and predictions, then compare disagreements.

Reveal six labeled creature cards, hide labels on four new cards, record each learner's rule and predictions, then compare disagreements.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

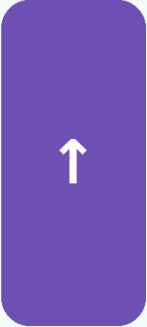
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

THIS WEEK'S CHECKPOINT Turn score thresholds into visible boundaries such as recommend / maybe / not now.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

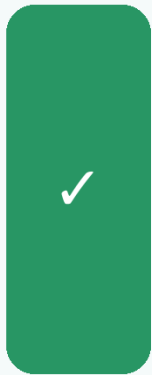
Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF

Train classmates on labeled creatures and compare their rules.

CHALLENGE Train classmates on labeled creatures and compare their rules.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Recommend if adventure AND pages<150; otherwise Other.

The question

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Worked explanation

A is Recommend. B fails length; C fails genre. Show both conditions.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Both conditions must pass

Evidence or stage · Value or result

Evidence or stage: Adventure, 120 pages

Value or result: Genre yes, length yes → Recommend

Evidence or stage: Adventure, 180 pages

Value or result: Genre yes, length no → Other

Evidence or stage: Science, 90 pages

Value or result: Genre no, length yes → Other

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

A is Recommend. B fails length; C fails genre. Show both conditions.

14. Try a new case independently

Independent case: Changed independent case

A written rule selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A human classifier uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what evidence is missing in the second case.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Recommend a book only if genre is science AND pages $\leq$ 150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.